

# PHASE-0 STUDENT ENGINEERING PACKET

*Behavioral-Intelligence MicroDrone Wildlife Deterrence System*

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## SECTION 1 — EXECUTIVE OVERVIEW

The Phase-0 MicroDrone Wildlife Deterrence System is a roof-mounted autonomous micro-drone designed to detect approaching birds, classify species, execute species-adaptive deterrence behaviors, and safely return to its dock. It uses contextual AI, multi-axis motion, dual-sensor detection, and strict safety layers to ensure effective deterrence without risk to humans, animals, or property.

Phase-0 is intentionally lightweight, safe, and student-buildable within one semester. It uses the NVIDIA Jetson Nano as the onboard compute module and incorporates a new **Interior Cart Deterrence Mode** to prevent habituation when birds enter the cart from the ground.

# SECTION 2 — FULL PHASE-0 SPECIFICATION

## 2.1 Core Concept

A micro-drone that detects wildlife, classifies species, selects deterrence patterns, executes multi-axis motion behaviors, and adapts based on context.

## 2.2 Dual-Sensor Detection Architecture

- **Front CSI camera:** species classification, approach-angle detection, multibird detection.
- **Rear CSI/USB camera:** threat direction, rotation direction, launch priority.
- **Interior bird-entry sensor:** detects birds entering the cart from the ground.

## 2.3 Multi-Axis Motion Engine

Exterior mode:

- Rise
- Tilt
- Rotation
- Hover
- Adaptive pitch/yaw/vertical acceleration
- Short-path rotation logic

Interior mode:

- No vertical rise
- Micro-yaw oscillation (5–15°)
- Micro-tilt pulses (3–8°)
- Directed LED/sound cues downward into cart
- Strict interior geofence

## 2.4 Species-Adaptive Intelligence Layer

Different species trigger different deterrence patterns:

- **Ravens:** aggressive, fast tilt, rapid escalation
- **Crows:** medium intensity, sweeping pattern
- **Grackles:** jittery, disruptive, fast escalation
- **Gulls:** high hover, circular pattern
- **Ibis/Herons:** gentle, slow, non-aggressive

Interior mode uses **scaled-down versions** of these patterns.

## 2.5 Safety Layer (Corrected)

Drone must ignore:

- Humans
- Dogs
- Golf carts
- Maintenance vehicles

Safety behaviors:

- No-launch zones
- No-tilt zones
- Controlled retreat
- Battery-aware behavior
- Docking abort logic
- Interior Cart Deterrence Mode
- Interior No-Rise Rule
- Human-Presence Rule

## 2.6 Docking & Re-Engagement Logic

- Magnetic alignment
- Charging contacts
- Docking abort
- Hover skip (low battery)
- Re-engagement triggers

## 2.7 Interior Cart Deterrence Mode

**If humans present inside the cart:**

- Drone stays docked
- Randomized LED strobe patterns
- Randomized buzzer patterns
- No fixed pattern (prevents habituation)

**If cart empty:**

- Drone stays docked
- Micro-motion deterrence:
  - 5–15° yaw oscillation
  - 3–8° tilt pulses

- No vertical rise
  - Strict interior geofence
- Directed LED/sound cues downward
- Species-adaptive modulation
- Randomized patterns

## 2.8 Enhancement Roadmap

- Species-adaptive deterrence patterns
- Species-adaptive escalation timing

## 2.9 Market Rationale

Large TAM (total addressable market) in golf-cart market; ideal for OEM and aftermarket.

# SECTION 3 — UPDATED BILL OF MATERIALS (WITH ACRONYM EXPLANATIONS)

### Core Compute

- Jetson Nano 2GB Developer Kit — onboard AI computer.

### Sensors

- CSI Cameras (Camera Serial Interface): high-speed cameras for real-time vision.
- IMU (Inertial Measurement Unit) via I<sup>2</sup>C bus: tilt/rotation/acceleration sensor.
- IR proximity sensor: short-range safety detection.
- Interior bird-entry sensor: detects birds entering cart.

### Motors & ESCs

- 4 × brushless motors (1103/1204)
- 4 × ESCs (Electronic Speed Controllers): convert battery power to motor power.

### Power

- LiPo 2S 500–800 mAh battery:
  - LiPo = Lithium-Polymer
  - 2S = 2 cells (~7.4V)
  - mAh = capacity rating
- 12V → 5V buck converter: powers Jetson and dock.

## **Deterrence Hardware**

- LED strobe
- Piezo buzzer

## **Docking Hardware**

- Magnetic alignment rails
- Spring-loaded charging contacts
- Dock alignment sensor
- Weather-resistant enclosure

## **Jetson Nano GPIO Connections**

- GPIO (General-Purpose Input/Output): programmable pins for motors, sensors, LEDs, buzzer.

# **SECTION 4 — WIRING & ELECTRONICS GUIDE**

## **Jetson → Sensors**

- CSI camera → CSI port
- Rear camera → USB or CSI
- IMU → I<sup>2</sup>C (SDA/SCL)
- IR sensor → GPIO 17
- Interior sensor → GPIO 23

## **Jetson → ESCs**

- ESC1 → GPIO 5
- ESC2 → GPIO 6
- ESC3 → GPIO 13
- ESC4 → GPIO 19
- ESC power → LiPo battery

## **Jetson → Dock**

- Dock sensor → GPIO 22
- Charging contacts → 5V rail

## **Jetson → Deterrence Hardware**

- LED → GPIO 24
- Buzzer → GPIO 25

## **Power**

- Jetson → 5V buck converter
- ESCs → LiPo battery
- Sensors → Jetson 5V rail

# **SECTION 5 — SYSTEM ARCHITECTURE DOCUMENT**

## **Detection Layer**

- Dual cameras
- Interior sensor
- IMU
- IR sensor

## **Decision Layer**

- SpeciesAdaptive Intelligence
- Motion parameter selection
- Deterrence pattern selection
- Interior mode logic
- Safety filtering
- Battery-aware behavior

## **Motion Layer**

- Exterior rise/tilt/rotation
- Interior micro-motion
- Adaptive rates
- Short-path rotation

## **Safety Layer**

- Human detection
- No-launch zones
- Interior cart mode
- Controlled retreat
- Docking abort

## **Docking Layer**

- Alignment
- Charging
- Abort
- Re-engagement

## **SECTION 6 — MOTION ENGINE SPECIFICATION**

### **Exterior Mode**

- Rise (1–3 feet)
- Tilt (predator posture)
- Rotation (full yaw)
- Hover
- Adaptive pitch/yaw/vertical acceleration
- Short-path rotation

### **Interior Mode**

- No vertical rise
- Micro-yaw oscillation (5–15°)
- Micro-tilt pulses (3–8°)
- Directed LED/sound cues
- Strict interior geofence
- Randomized patterns
- Species-adaptive modulation

## **SECTION 7 — SPECIES-ADAPTIVE INTELLIGENCE SPECIFICATION**

### **Ravens:**

Fast, aggressive, deep tilt, rapid escalation.

### **Crows:**

Medium intensity, sweeping pattern.

### **Grackles:**

Jittery, disruptive, fast escalation.

**Gulls:**

High hover, circular pattern.

**Ibis/Herons:**

Gentle, slow, non-aggressive.

**Interior Mode Constraints:**

Scaled-down versions with randomized patterns.

## **SECTION 8 — SAFETY LAYER SPECIFICATION**

- Ignore humans, dogs, carts
- No-launch zones
- No-tilt zones
- Controlled retreat
- Battery-aware behavior
- Docking abort
- Interior No-Rise Rule
- Human-Presence Rule
- Interior Motion Geofence
- Habituation prevention (randomized patterns)

## **SECTION 9 — DOCKING SYSTEM SPECIFICATION**

- Magnetic alignment
- Charging contacts
- Dock sensor
- Dock abort
- Hover skip (low battery)
- Re-engagement triggers

## **SECTION 10 — INTERIOR CART DETERRENCE MODE SPECIFICATION**

**Trigger:**

Interior sensor detects bird inside cart.

**If humans present:**

- No flight
- Randomized LED/buzzer patterns

**If cart empty:**

- Drone remains docked
- Micro-motion deterrence
- 5–15° yaw oscillation
- 3–8° tilt pulses
- No vertical rise
- Directed LED/sound cues
- Species-adaptive modulation
- Randomized patterns

## **SECTION 11 — BUILD PLAN (WEEK-BY-WEEK)**

**Week 1: Jetson setup, environment, libraries**

**Week 2: Camera + IMU + interior sensor integration**

**Week 3: ESC + motor wiring, bench tests**

**Week 4: Motion engine implementation**

**Week 5: Species-adaptive logic**

**Week 6: Safety layer + interior mode**

**Week 7: Dock construction + alignment logic**

**Week 8: Full system integration**

**Week 9: Field testing**

**Week 10: Demo preparation**

## SECTION 12 — TESTING & VALIDATION PLAN

- Bench tests
- Sensor tests
- Motion tests
- Dock tests
- Safety tests
- Interior mode tests
- Species-adaptive tests
- Full system tests

## SECTION 13 — GLOSSARY OF TERMS & ACRONYMS

- **CSI:** Camera Serial Interface
- **IMU:** Inertial Measurement Unit
- **PC:** Two-wire sensor communication bus
- **ESC:** Electronic Speed Controller
- **LiPo:** Lithium-Polymer battery
- **2S:** Two LiPo cells in series (~7.4V)
- **mAh:** Battery capacity
- **GPIO:** General-Purpose Input/Output pins